

Jon Arrizabalaga

Postdoctoral Researcher at Massachusetts Institute of Technology

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Education

PhD	Technical University of Munich , Robotics & Control • Thesis , Defense	2020 – 2025
MSc	KTH Royal Institute of Technology , Mechatronics Engineering • Thesis	2018 – 2020
BSc	University of Navarre , Mechanical Engineering	2014 – 2018

Academic Experience

Massachusetts Institute of Technology , Postdoctoral Researcher	2026 – present
Carnegie Mellon University , Visiting PhD / Postdoctoral Researcher	2024 – 2026
Charles University , Visiting PhD	2023
Shanghai Jiao Tong University , Visiting Student	2017 – 2018

Industry Experience

Bosch Research , MSc. Thesis - Robotics Researcher • Paper , Video	2020
Porsche AG , Research Intern	2019

Honors & Awards

Best Robocup Paper Award (finalist) 5 nominations out of ~4000 submissions at the conference IEEE/RJS IROS • Paper , Talk	2024
Best PhD Lecturer Award Recognizes the PhD lecturer who has received the highest evaluations from BSc and MSc students	2022
Outstanding End-of-Degree Award (finalist) Honors BSc. students who have excelled in both their academic and extracurricular endeavors	2018

Invited Talks

Northeast Systems and Control Workshop , Columbia University	2025
Talking Robotics , Virtual Seminar • Talk	2025
GRASP Laboratory , Vijay Kumar Lab, University of Pennsylvania	2024
German Aerospace Center (DLR) , Institute of Flight Systems	2023
Robotics and Perception Group , University of Zurich	2022

Teaching

Engineering Mechanics I , Lecturer, Technical University of Munich <i>Level: BSc, Students: 200 ~ 450</i>	2021 – 2024
Mechanics for Aerospace , Lecturer, Technical University of Munich (Asia) <i>Level: MSc, Students: 20 ~ 25</i>	2021 – 2023
Introduction to ROS (Robot Operating System) , Lecturer, Technical University of Munich <i>Level: MSc, Students: 50</i>	2021 – 2022
Dynamics Motion and Control , Teaching Assistant, KTH Royal Institute of Technology <i>Level: MSc, Students: 50</i>	2019 – 2020
Robust Mechatronics , Teaching Assistant, KTH Royal Institute of Technology <i>Level: MSc, Students: 50</i>	2019 – 2020

Publications

A Differentiable Interior-Point Method in Single Precision • Paper , Code , Website <i>J. Arrizabalaga, K. Tracy, Z. Manchester, Under review</i>	2026
Implicit Primal-Dual Interior-Point Methods for Quadratic Programming • Paper , Code <i>J. Arrizabalaga, Z. Manchester, Under review</i>	2026
ADMM-based Continuous Trajectory Optimization in Graphs of Convex Sets • Paper <i>L. Pries, J. Arrizabalaga, Z. Manchester, M. Ryll, Under review</i>	2026
TinySDP: Real Time Semidefinite Optimization for Certifiable and Agile Edge Robotics • Paper , Code , Website <i>I. Mahajan, J. Arrizabalaga, A. Grillo, F. Vega, J. Anderson, Z. Manchester, B. Plancher, RSS</i>	2026
DQ-NMPC: Dual-Quaternion NMPC for Quadrotor Flight • Paper , Video , Code , Website <i>L.F. Recalde, D. Agrawal, J. Arrizabalaga, G. Li, RA-L</i>	2025
Convex Maneuver Planning for Spacecraft Collision Avoidance • Paper , Code <i>F. Vega, J. Arrizabalaga, R. Watson, Z. Manchester, IEEE iSpaRo</i>	2025
The Trajectory Bundle Method: Unifying Sequential-Convex Programming and Sampling-Based Trajectory Optimization • Paper <i>K. Tracy, J. Zhang, J. Arrizabalaga, S.Schaal, Y. Tassa, T. Erez, Z. Manchester, IEEE ICRA</i>	2025
Conflict-Based Search as a Protocol: A Multi-Agent Motion Planning Protocol for Heterogeneous Agents, Solvers, and Independent Tasks • Paper , Website <i>R. Veerapaneni,, A. Tang, H. He, S. Zhao, V. Shah, Y. Cen, Z. Ji, G. Olin, J. Arrizabalaga, Y. Shaoul, M. Likhachev Likhachev, IEEE ICRA</i>	2025
A Universal Formulation for Path-Parametric Planning and Control • Paper , Code , Website , Talk <i>J. Arrizabalaga, Z. Sir, Z. Manchester, M. Ryll, IEEE CSM</i>	2025
PHODCOS: Pythagorean Hodograph-based Differentiable Coordinate System • Paper , Code , Talk <i>J. Arrizabalaga, F. Vega, Z. Sir, Z. Manchester, M. Ryll, IEEE Aero</i>	2025
Differentiable Collision-Free Parametric Corridors • <i>Best Paper Finalist</i> • Paper , Code , Video , Talk <i>J. Arrizabalaga, Z. Manchester, M. Ryll, IEEE IROS</i>	2024
Geometric Sloss-Free Tracking for Robotic Manipulators • Paper , Code , Video , Talk <i>J. Arrizabalaga, L. Pries, R. Laha, R. Li, S. Haddadin, M. Ryll, IEEE ICRA</i>	2024
Learning for CasADi: Data-Driven Models in Numerical Optimization • Paper , Code , Talk <i>T. Salzmann, J. Arrizabalaga, J. Andersson, M. Pavone, M. Ryll, L4DC</i>	2023
Pose-Following with Dual Quaternions • Paper , Code , Talk <i>J. Arrizabalaga, M. Ryll, IEEE CDC</i>	2023
SCTOMP: Spatially Constrained Time-Optimal Motion Planning • Paper , Video <i>J. Arrizabalaga, M. Ryll, IEEE IROS</i>	2023
Spatial Motion Planning with Pythagorean Hodograph curves • Paper , Video , Talk <i>J. Arrizabalaga, M. Ryll, IEEE CDC</i>	2022
Towards Time-Optimal Tunnel-Following for Quadrotors • Paper , Video <i>J. Arrizabalaga, M. Ryll, IEEE ICRA</i>	2022
Real-Time Neural-MPC: Deep Learning Model Predictive Control for Quadrotors and Agile Robotic Platforms • Paper , Code <i>T. Salzmann, E. Kaufmann, J. Arrizabalaga, M. Pavone, D. Scaramuzza, M. Ryll, RA-L</i>	2022
A caster-wheel-aware MPC-based motion planner for mobile robotics • Paper , Thesis , Video <i>J. Arrizabalaga, N. van Duijkeren, M. Ryll, R. Lange, IEEE ICAR</i>	2021

Students Mentored

Viraj Shah , Research Project, <i>now at</i> Carnegie Mellon University	2025
Eden Cen , Research Project, <i>now at</i> Carnegie Mellon University	2025

Akshay Jaitly , Research Project, <i>now at</i> Worcester Polytechnic Institute	2025
Luis Fernando Recalde , Research Project, <i>now at</i> Worcester Polytechnic Institute	2024
Giuseppe Soldati , Semester Thesis, <i>now at</i> German Aerospace Center (DLR)	2023
Harun Tongay Tamtürk , Semester Thesis & MSc. Thesis, <i>now at</i> Airbus Defence and Space	2023
Bruno Sorban , Research Project, <i>now at</i> Boston Consulting Group	2023
Tommaso Faraci , Research Project, <i>now at</i> Università di Trento	2022

Extracurricular Projects

Ascent - Rocket Hopper , Guidance and Control Lead • Website	2022 – 2024
The Juggling Robot Project , Team Lead • Video	2019 – 2020
Formula Student Electric , Vehicle Dynamics Engineer	2016 – 2018

Review Activities

- IEEE Transactions on Robotics (T-RO)
- IEEE Robotics and Automation Letters (RA-L)
- IEEE Transactions on Systems, Man, and Cybernetics: Systems (T-SMC)
- Robotics Science and Systems (RSS)
- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- IEEE Conference on Decision and Control (CDC)
- IEEE International Conference on Humanoid Robots (Humanoids)
- IEEE International Conference on Space Robotics (iSpaRo)